A Computational Framework for Optimal Democratic Governance

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Abstract

This paper introduces the Pareto Legislative Engine (PLE) and Adaptive Trust Networks (ATN), a unified computational framework that redefines democratic governance as an optimization problem. By combining quadratic voting, machine learning–based policy simulation, and trust-weighted decision systems, this model addresses systemic issues such as polarization, inefficiency, and lack of transparency.

Introduction

Modern democratic systems rely heavily on majority voting, which captures preference direction but fails to measure intensity. This leads to suboptimal decisions and persistent gridlock. This research reframes governance as a measurable system that can be optimized under constraints, similar to problems in economics and engineering.

Pareto Legislative Engine

The PLE operationalizes mechanism design at scale. It uses quadratic voting to capture preference intensity, retrieval-augmented generation to simulate real-world policy impacts, and multi-agent optimization to identify Pareto-efficient legislative solutions. This transforms policymaking into a structured, data-driven process.

Adaptive Trust Networks

The ATN introduces a dynamic credibility framework that assigns trust scores to participants based on accuracy, expertise, and consistency. This allows decisions to incorporate both public preferences and reliable knowledge, reducing misinformation and improving robustness.

System Integration

Together, PLE and ATN form a three-layer system: preference measurement, credibility weighting, and optimization. This architecture creates a closed-loop system capable of continuously improving governance outcomes.

Implications

This framework extends mechanism design into large-scale political systems and offers a pathway toward reducing polarization. By shifting incentives from conflict to optimization, it provides a scalable foundation for stable global governance.

Conclusion

This work demonstrates that democratic governance can be transformed through computational methods. By integrating mathematics, AI, and economics, it presents a new paradigm for transparent, efficient, and trustworthy decision-making.